

15.439 Quantitative Investment Management

Assignment 1

February 24, 2026

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1 Abstract / Executive memo

Data quality: The WKKWNT 4sight sample (1,045 rows) has several critical and medium-severity errors: one monotonic date violation (row 404: 2017-06-21 after 2020-06-20), two negative/zero prices (2018-06-06, 2018-10-10), three duplicate dates (2016-08-19, 2016-10-31, 2016-11-03; 2016-10-31 has conflicting content), 23 OHLC bracket violations, 7 signal sentinels (-999 , 6×0), and 1 null in Low. Raw data was not modified; all corrections are applied in code and documented in a corrections ledger.

Signal effectiveness: After applying Q1 corrections, the 4sight signal **does** show statistically significant predictability, but only for specific uses. **Strongest use: forward volatility.** The signal reliably predicts realized volatility over 5–60 days (best: 60d, IC ≈ 0.34 with 5-day moving average of signal). It is well suited for volatility targeting, position sizing, or hedging. **Moderate use: 20-day return direction.** At 20-day horizon, the signal helps predict direction (Rank IC ≈ -0.18 , hit rate $\approx 62\%$) but not return level. One-day and return-level prediction are not supported. **Use level or smoothed level** of the signal; percentage change and sign of the signal add little.

Recommendation: Use the signal for forward-volatility prediction (20d/60d) and for medium-horizon (20d) directional views. Do not rely on it for next-day returns or return magnitude. Apply the corrections in analysis code as implemented.

2 Initial analysis

2.1 Methodology

The initial analysis was run by a single pipeline (`scripts/01_initial_analysis.py`) that loads the raw WKKWNT CSV, normalizes column names (Date, Signal, Open, High, Low, Close, Adj Close $\rightarrow$ date, signal, open, high, low, price, adj_close), and executes 17 checks in sequence. Raw data is never modified; findings and proposed corrections are recorded separately. Outputs a normalized copy with optional OHLC boolean columns and figures/counts per check.

Checks performed: (1) null counts per column; (2) duplicate calendar dates; (3) duplicate-content audit for those dates; (4) monotonicity of dates; (5) date coverage vs. business-day calendar; (6) weekend/holiday flags; (7) negative/zero prices (Adj Close ≤ 0); (8) OHLC bracket (high $\geq$ low, open/close in [low, high]); (9) Adj Close vs Close (diagnostic only); (10) return outliers ($|r| > 20\%$); (11) signal sentinels (-999 , zeros); (12) signal z-score; (13) signal day-over-day jumps; (14) corporate-action ratio (Adj Close/Close) jumps; (15) schema/dtypes; (16) date gaps > 5 days; (17) issue summary counts.

2.2 Errors found

All errors identified are listed in Table 1 in the Appendix. Summary:

Critical: Monotonic date violation (row 404: 2017-06-21 immediately after 2020-06-20); two negative/zero prices (2018-06-06 adj_close = 0, 2018-10-10 adj_close = -152.28). These break time ordering and return calculations.

Medium: Duplicate dates (3 dates, 6 rows): 2016-08-19 and 2016-11-03 identical copies (keep first); 2016-10-31 has conflicting OHLC/signal (escalate to vendor). OHLC bracket: 3 high < low, 3 open outside [low, high], 17 close outside [low, high] (23 total). Signal sentinels: 1×-999 , 6×0 (treat as missing).

Low / informational: 1 null in Low; 9 weekend dates; 2 large gaps (10d, 166d; 166d linked to monotonicity). Return outliers (15) and extreme signal z-score/jumps are symptoms of the bad prices and sentinels; they are

resolved by the proposed price and signal corrections. $\text{Adj Close} \neq \text{Close}$ is expected (adjustments); not an error.

3 Signal analysis

3.1 Methodology

Analysis uses an *analysis-ready* dataset built by applying Q1 corrections in code: sort by date; deduplicate dates (keep first); fix monotonicity (drop row causing backward step); drop rows with $\text{adj_close} \leq 0$; replace signal in $\{-999, 0\}$ with NaN. No lookahead: forward return at t is $(P_{t+1}/P_t) - 1$; forward vol over H days is the standard deviation of daily returns in $[t + 1, t + H]$.

Targets: Forward returns at horizons 1, 5, 20. Forward-looking realized volatility over 5, 20, 60 days.

Predictors: Raw signal; 5/20-day backward-looking rolling mean (`signal_ma_5`, `signal_ma_20`); signal percentage change (`signal_pct`); sign of signal (`sign_signal`).

Metrics (four per target $\times$ predictor): (1) IC (Pearson) with Fisher z -test, $H_0: \rho = 0$. (2) Hit rate (directional accuracy) with binomial test, $H_0: p = 0.5$. (3) OLS regression of target on predictor; t -test on slope β . (4) Rank IC (Spearman) with Fisher z -test. All use pairwise dropna; CIs and p -values reported.

Pipeline: `scripts/02_signal_effectiveness.py` builds the analysis-ready frame, adds multi-horizon targets and predictor columns, then loops over all target $\times$ predictor pairs and computes the four metrics. Outputs: `results/q2_metrics.csv` (long format), `md_files/Q2_analysis.md`.

3.2 Findings

Forward volatility IC is positive and significant for all level-based predictors and horizons 5, 20, 60. Best: `forward_vol_60  $\times$  signal_ma_5` (IC = 0.34, $p \approx 0$). 20d vol: IC 0.16–0.20; 5d vol: IC 0.09–0.10. Higher signal (or its moving average) predicts higher subsequent realized vol. Hit rate for vol is ≈ 99 –100% (same sign). Regression slopes are large and significant.

Return level: IC and β are small (≈ 0.04) and not significant at any horizon. The signal does not meaningfully predict return *magnitude*.

Return direction: At 20d horizon, Rank IC ≈ -0.12 to -0.18 ($p < 0.001$), hit rate $\approx 62\%$ ($p \approx 0$). At 5d, `signal_ma_20` has significant Rank IC (-0.08 , $p \approx 0.01$). At 1d, only weak hit-rate significance ($\approx 55\%$); not reliable for next-day direction.

Predictor choice: `signal_pct` and `sign_signal` are uninformative (ICs and Rank ICs near zero, rarely significant). Level or smoothed level (raw signal, `signal_ma_5`, `signal_ma_20`) drives the results.

A compact summary of metrics by target and predictor is in Table 2 in the Appendix.

Appendix

3.3 Table of errors found

Table 1: Errors found in WKKWNT sample and proposed action.

#	Issue type	Count / key	Proposed action
1	Null (low)	1 row	Impute or drop; log row/date
2	Duplicate dates	3 dates, 6 rows	Keep first for 2; escalate 2016-10-31
3	Monotonic date	Row 404 (2017-06-21 after 2020-06-20)	Re-sort or drop misplaced row
4	Negative/zero price	2 rows (2018-06-06, 2018-10-10)	Drop or NaN; do not use for returns
5	OHLC bracket	3 high<low; 3 open; 17 close	Log; exclude or request correction
6	Return outliers	15 rows	Resolved by fixing bad prices
7	Signal sentinels	1 $\times$ -999, 6 $\times$ 0	Treat as missing (NaN)
8	Weekend dates	9	Log; optionally exclude
9	Gaps	10d, 166d	Re-check after monotonicity fix

3.4 Table of signals analysed

Summary of targets and predictors.

Table 2: Targets and predictors used in signal analysis.

Target	Description	Predictors
forward_return_1	1-day forward return	signal, signal_ma_5, signal_ma_20, signal_pct, sign_signal
forward_return_5	5-day forward return	(same)
forward_return_20	20-day forward return	(same)
forward_vol_5	5-day forward realized vol	(same)
forward_vol_20	20-day forward realized vol	(same)
forward_vol_60	60-day forward realized vol	(same)

Metrics per cell: IC, Rank IC, hit rate, OLS β ; each with p -value and 95% CI.

Full results table. Table 3 reports IC, $p(\text{IC})$, Rank IC, $p(\text{Rank IC})$, hit rate, $p(\text{HR})$, β , $p(\text{reg})$ for each target $\times$ predictor. $p < 0.05$ indicates significance.

Table 3: Full signal analysis results: IC, Rank IC, hit rate, OLS β and p -values by target and predictor.

Target	Predictor	IC	$p(\text{IC})$	Rank IC	$p(\text{Rk})$	Hit	$p(\text{HR})$	β	$p(\beta)$
<i>Return horizon 1</i>									
ret 1	signal	0.039	0.206	0.019	0.535	0.548	0.003	9.67	0.206
	signal_ma_5	0.044	0.156	0.016	0.610	0.549	0.002	12.57	0.156
	signal_ma_20	0.044	0.153	-0.028	0.364	0.549	0.002	13.49	0.153
	signal_pct	-0.001	0.985	-0.029	0.353	0.492	0.615	-0.16	0.985
	sign_signal	0.001	0.975	-0.020	0.512	0.548	0.003	12.88	0.975
<i>Return horizon 5</i>									
ret 5	signal	0.039	0.207	0.033	0.283	0.562	0.000	9.67	0.207
	signal_ma_5	0.044	0.155	-0.021	0.508	0.561	0.000	12.66	0.155
	signal_ma_20	0.045	0.151	-0.080	0.010	0.561	0.000	13.69	0.151
	signal_pct	-0.001	0.985	0.023	0.464	0.503	0.851	-0.16	0.985
	sign_signal	0.001	0.975	-0.038	0.224	0.562	0.000	12.93	0.975
<i>Return horizon 20</i>									
ret 20	signal	0.040	0.201	-0.115	0.000	0.623	0.000	10.15	0.201
	signal_ma_5	0.045	0.148	-0.160	0.000	0.624	0.000	13.42	0.148
	signal_ma_20	0.046	0.144	-0.179	0.000	0.624	0.000	14.50	0.144
	signal_pct	-0.001	0.985	0.020	0.518	0.510	0.529	-0.16	0.985
	sign_signal	0.001	0.975	-0.041	0.192	0.623	0.000	13.36	0.975
<i>Vol horizon 5</i>									
vol 5	signal	0.085	0.006	-0.195	0.000	0.999	0.000	20.85	0.006
	signal_ma_5	0.096	0.002	-0.174	0.000	1.000	0.000	27.35	0.002
	signal_ma_20	0.099	0.001	-0.087	0.005	1.000	0.000	30.26	0.001
	signal_pct	-0.002	0.945	-0.002	0.957	0.506	0.730	-0.57	0.945
	sign_signal	0.002	0.944	0.017	0.584	0.999	0.000	28.80	0.944
<i>Vol horizon 20</i>									
vol 20	signal	0.163	0.000	-0.167	0.000	0.999	0.000	39.90	0.000
	signal_ma_5	0.189	0.000	-0.125	0.000	1.000	0.000	54.20	0.000
	signal_ma_20	0.202	0.000	-0.069	0.028	1.000	0.000	61.80	0.000
	signal_pct	-0.005	0.874	-0.017	0.595	0.507	0.683	-1.30	0.874
	sign_signal	0.005	0.887	0.051	0.106	0.999	0.000	58.34	0.887
<i>Vol horizon 60</i>									
vol 60	signal	0.291	0.000	0.059	0.065	0.999	0.000	70.39	0.000
	signal_ma_5	0.342	0.000	0.105	0.001	1.000	0.000	97.87	0.000
	signal_ma_20	0.335	0.000	0.110	0.001	1.000	0.000	102.56	0.000
	signal_pct	-0.009	0.781	-0.001	0.966	0.504	0.822	-2.25	0.781
	sign_signal	0.008	0.799	0.055	0.087	0.999	0.000	103.42	0.799

3.5 Relevant code snippets

3.5.1 Building the analysis-ready dataset (Q1 corrections)

From `src/clean_for_analysis.py`: core cleaning and forward-return construction.

Listing 1: `build_analysis_ready_df` and `add_forward_return`

```

1 def build_analysis_ready_df(raw_df_or_path):
2     # ... load + normalize ...
3     df = df.sort_values(date_col).reset_index(drop=True)
4     df = df.drop_duplicates(subset=[date_col], keep="first")
5     # Fix monotonicity: drop row causing backward date step
6     ser = pd.to_datetime(df[date_col], errors="coerce")
7     backward = ser.diff() < pd.Timedelta(0)
8     if backward.any():
9         df = df.drop(index=df.index[backward]).reset_index(drop=True)
10    # Drop bad prices
11    df = df[df[adj_col] > 0].reset_index(drop=True)
12    # Signal sentinels -> NaN
13    df.loc[df[signal_col].isin([-999, 0]), signal_col] = np.nan
14    return df
15
16 def add_forward_return(df):
17     out = df.copy()
18     out["forward_return"] = out[adj_col].pct_change().shift(-1)
19     return out

```

3.5.2 Information coefficient with Fisher z -test

From `src/metrics_signal.py`: Pearson IC and Fisher z for $H_0: \rho = 0$.

Listing 2: `compute_ic`

```

1 def compute_ic(signal, forward_return):
2     s, r = _dropna_pair(signal, forward_return)
3     n = len(s)
4     if n < 4: return np.nan, n, np.nan, np.nan, np.nan
5     ic = np.corrcoef(s, r)[0, 1]
6     z, se = _fisher_z(ic, n) # z = 0.5*ln((1+r)/(1-r)), se = 1/sqrt(n-3)
7     z_stat = z / se
8     p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))
9     ci_low, ci_high = _z_to_r(z - 1.96*se), _z_to_r(z + 1.96*se)
10    return float(ic), n, float(p_value), float(ci_low), float(ci_high)

```

3.5.3 Runner: targets and predictors loop

From `scripts/02_signal_effectiveness.py`: configuration and main loop (abbreviated).

Listing 3: Target/predictor config and loop

```

1 TARGET_COLS = [
2     ("return", 1, "forward_return_1"),
3     ("return", 5, "forward_return_5"),
4     ("return", 20, "forward_return_20"),
5     ("vol", 5, "forward_vol_5"),
6     ("vol", 20, "forward_vol_20"),
7     ("vol", 60, "forward_vol_60"),
8 ]
9 PREDICTOR_COLS = [
10    ("signal", None, "signal"),
11    ("signal_ma", 5, "signal_ma_5"),
12    ("signal_ma", 20, "signal_ma_20"),
13    ("signal_pct", None, "signal_pct"),

```

```
14     ("sign_signal", None, "sign_signal"),
15 ]
16 # ...
17 for (ttype, thORIZ, tcol) in TARGET_COLS:
18     for (ptype, pparam, pcol) in PREDICTOR_COLS:
19         pred, target = df[pcol], df[tcol]
20         for method_name, method_fn, ... in METHODS:
21             # call method_fn(pred, target), append row with target_col, predictor_col,
                method, estimate, n, p_value, ...
```